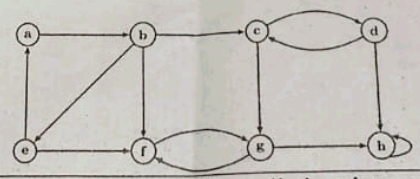
Roll Number.....

Department of Computer Engineering
National Institute of Technology Kurukshetra-136119
Mid Term- I, Odd Semester, September, 2025
Subject: Design and Analysis of Algorithms (ITPC-201/AIPC-203)
Branch: B.Tech. IT, AIML (3rd Semester)

Maximum Marks: 15 **Time:** 50 Minutes

Instructions: All questions are compulsory.

Q.1)	a). Solve the following recurrence relation: $T(n)=2T(n-1)+n2^n$ for $n > 0$, $T(0)=1$.	[2+3]
	b). Given a hash table with size=11 entries and the following hash function, h_1(key)=key % size and step function, h_2(key) = 1+{key % 7}. Insert the keys {63, 50, 25, 79, 67, 24} in the given order (from left to right) to hash table using double hashing with h_1 as hash function and h_2 as step function. Determine the final hash table, number of collisions for each element, and efficiency of the hashing scheme for given data.	
Q.2)	For the given graph show how DFS works starting from node 'C'. Assume that the procedure considers the vertices in the alphabetic order and adjacency list is ordered alphabetically. Show the discovery time and finish time of each vertex. Determine the classification of each edge. Also, represent the traversal using parenthesis structure. 	[5]
Q.3)	For the given an unsorted array, A= A[1…n] containing n distinct integers, design an algorithm that outputs the smallest k elements in the array A. The time complexity of the designed algorithm should be O(n+klogn). Write the pseudo code and discuss time complexity of each step.	[5]